Google Data Analytics

Capstone Project

SQL Retail Case Study

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March 2026

The purpose of this case study is two-fold: (1) Illustrate skills using SQL and (2) Illustrate a practical application of the Google Data Analytics six step framework. For this capstone project. The data source for this project is pulled from Kaggle and authored by Hassan Jameel Ahmed. MySQL was used for the data analysis. Excel was used for data compilation.

Source: https://www.kaggle.com/datasets/hassanjameelahmed/store-sales/data

Ask

The audience for the findings from the analysis from this capstone would be store management and executive management. The problem that is trying to be solved is to better understand customers and identify trends in various parts of the database. Knowing this new information will help enable wiser decisions to be made regarding buying and selling certain merchandise.

Prepare

Data is located in Kaggle.
The data is organized with the following columns: Customer, Age, Gender, Category, Item Purchased, Amount, Season, Payment Method, Item Rating, Discount Applied, and Previous Purchases.
The dataset is reliable, original, comprehensive, current, and cited.
This data is a public dataset with no licensing, privacy, or security concerns.

Process

To ensure data cleanliness I eyeballed the data for issues and also checked for spelling differences using filtering, and major outliers in the data.

Analyze

To analyze the data, I used SQL. In this section I will share the business question, coding that was used, and results when relevant.

1. ***How can I view all records of the data?***

```
SELECT *
FROM sqlretail3
```

All 5000 records are illustrated.

2. What are the top five items for each season by gender?

```
SELECT *
FROM sqlretail3
WHERE Season='Autumn' AND GENDER='Male'
ORDER BY AMOUNT DESC;
```

(Season and gender were changed for data collection)

	Autumn	Winter	Spring	Summer
Female	Lamp x4, sneakers	Curtains, Lamp, kitchen set, Dumbells, formal shoes	Curtains x3, Kitchen set, sneakers	Curtains x3, kitchen set, lamp
Male	Electronics x5	Electronics x5	Electronics x5	Electronics x5

Electronics were leading with purchases for male customers year round. Females had a little bit more variation, but lamps and curtains were common throughout the year with top purchases.

3. How many unique items were purchased?

```
SELECT COUNT(DISTINCT ItemPurchased) AS Total_Items
FROM sqlretail3
```

Total items=30

4. ***I would like to view customer 687's results.***

```
SELECT *
FROM sqlretail3
WHERE CustomerID=687
```

Partial data shown for illustration purposes:

CustomerID	Age	Gender	Category	ItemPurchased	Amount	Season
687	20	Female	Accessories	Wallet	96.93	Spring

5. Can you send me just the data for accessories?

```
SELECT *
FROM sqlretail3
WHERE Category='Accessories'
```

6. *Does previous sales influence purchase?*

```
SELECT COUNT(*) AS "Quantity"
FROM sqlretail3
WHERE PreviousPurchases=3;
```

(Number was changed and adapted)

Count and percentage by previous purchases

	Count	Percentage
0	33	0.66%
1	157	3.14%
2	411	8.22%
3	683	13.66%
4	890	17.80%
5	895	17.90%
6	763	15.26%
7	514	10.28%
8	319	6.38%
9	179	3.58%
10	98	1.96%
11	38	0.76%
12	16	0.32%
13	4	0.08%

Very few customers were first time customers. Over half of the customers purchased 3-7 times before.

7. I would like to see a snapshot of the data, but only a few rows to get a sense of the data.

```
SELECT *
FROM sqlretail3
LIMIT 3
```

8.How are the customers grouped by age?

<20 Quantity	20 to 29 Quantity	30 to 39 Quantity	40 to 49 Quantity	50 to 59 Quantity	60 to 69 Quantity	70 to 79 Quantity	80 to 89 Quantity	90 to 99 Quantity
0	938	946	1039	1014	965	98	0	0
0.00%	18.76%	18.92%	20.78%	20.28%	19.30%	1.96%	0.00%	0.00%

Most of the customers tend to be 20-69 in age.

9. ***I'm interested in the count of any items that had "all" to see if there may be a trend relating to that.***

```
SELECT COUNT(*) AS Count
FROM sqlretail3
WHERE ItemPurchased
LIKE '%ALL%'
```

Only 445 customers or 8.9%. Thus, simply having "all" in the item name didn't result in many purchases.

10. ***What were the top categories purchased?***

```
SELECT Category, SUM(AMOUNT) AS Count
FROM sqlretail3
GROUP BY Category
ORDER BY COUNT DESC;
```

Category	Amount	Percentage of Sales
Electronics	895402.35	62.82%
Footwear	141142.24	9.90%
Sports	115598.11	8.11%
Womens Clothing	61718.41	4.33%
Mens Clothing	59680.43	4.19%
Accessories	47633.83	3.34%
Home	46374.22	3.25%
Beauty	43887.06	3.08%
Groceries	14015.96	0.98%

Electronics, footwear, and sports were the top categories.

11. ***What were the top items sold?***

```
SELECT ItemPurchased, COUNT(itempurchased) AS Count
FROM sqlretail3
GROUP BY itempurchased
ORDER BY COUNT DESC;
```

Item	Count
Sandals	339
Formal Shoes	324
Sneakers	320
Yoga Mat	283
Dumbbells	277

12. ***What are the top maximum amount spent per item? Minimum? Average?***

```
SELECT Itempurchased, MAX(Amount) AS Amount
FROM sqlretail3
GROUP BY ItemPurchased
ORDER BY Amount DESC
```

Maximum item

Item	Dollars
Laptop	2997.94
Headphones	2995.9
Smart Watch	2994.75

Minimum items

Item	Dollars
Beverages	5.08
Rice Pack	5.44
Cooking Oil	5.85

Average

```
SELECT AVG(Amount) AS Average
FROM sqlretail3
```

$285.09 is the average spent per customer.

13. ***How many are female vs. male?***
SELECT COUNT(GENDER) as COUNT
FROM sqlretail3
WHERE Gender='female'

Female- 2503
Male-2496

Roughly 50% for each gender.

14. ***How many transactions were card vs. cash?***
SELECT Paymentmethod,
COUNT(*) AS Paymentcount
FROM sqlretail3
GROUP BY PaymentMethod
ORDER BY Paymentcount DESC;

Payment Method	Count
Card	4009
Cash on Delivery	991

More transactions were found using card.

15.*What is the average item rating?*
SELECT AVG (ItemRating) AS "Item Average"
FROM sqlretail3;
3.78

Share and Act

1 Page Summary of Findings and Recommendations
Retail Store Sales Data

By the Numbers

- **5000** Records
- **~50%** Female and **50%** Male
- **98%** of customers aged 20-69
- **30** unique Items
- **9** Categories
- **~15%** average discount*
- **3.78** Average Rating

*- Found in Excel

Key Findings

- Men purchased electronics consistently throughout the year
- Women tended to purchase lamps and curtains
- Card was used for 80% of the purchases
- Over 50% of the customers purchased 3-7 times before
- Footwear items (sandals, formal shoes, and sneakers) were the most plentiful in regards to amount sold
- Electronics accounted for 62.82% of sales followed by footwear with 9.9%

Recommendations

- Promote electronics and footwear more to build on previous sales
- Reduce money spent on groceries
- Consider specializing in the top three categories (Electronics, footwear, and sports)
- Ensure card readers are working and up to date as 80% of transactions use card

Future Analysis

- What is the reason for having an average discount of 15%?
- Why were some items rated lower than others?
- What would be the threshold for ordering lower rating items vs. considering investing in more higher rated items?